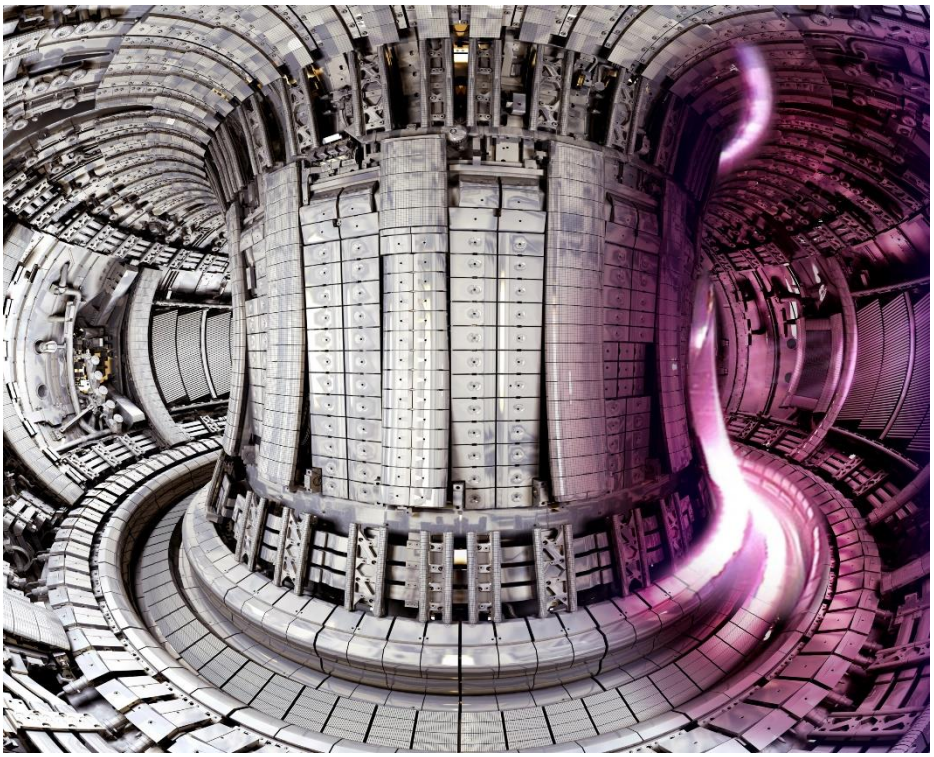
IMPERIAL

A Review of the Science, Engineering, Technology and Design of Nuclear Fusion Blankets

ME3 Literature Research Project



Interior of JET with its new ITER-like wall of tungsten and beryllium; credit: UKAEA

[1]

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[1] https://euro-fusion.org/devices/jet/

Abstract

Breeder blankets are integral to achieving sustainable fusion power, serving as tritium breeders, thermal energy extractors and neutron shields within fusion reactors. This report reviews the evolution of breeder blanket designs, from early theoretical concepts to refined configurations from programs across the world, such as EU DEMO, ITER and CFETR. It aims to critically evaluate their current state of research and development. The review examines solid breeder concepts such as the Helium-Cooled Pebble Bed (HCPB) and Water-Cooled Ceramic Breeder (WCCB), alongside liquid metal approaches like the Dual-Coolant Lead-Lithium (DCLL) blanket.

Emphasis is placed on engineering trade-offs related to tritium breeding ratio, heat removal efficiency, irradiation tolerance and other design-specific effects such as corrosion resistance or the magnetohydrodynamic (MHD) effect. This review highlights the integration of advanced multiscale and multiphysics modelling techniques with neutronic simulations and data from emerging experimental facilities to refine and guide design improvements.

Key findings indicate that, while high tritium breeding ratio and thermal efficiencies are achievable, significant R&D gaps persist in material compatibility, manufacturing techniques and tritium extraction methods. Ongoing and future efforts focusing on improved materials, refined computational frameworks and enhanced experimental test rigs will be crucial to advancing breeder blanket technology toward reliable, commercial-scale fusion energy.

Objectives

The following are the objectives of this paper.

Timeline of progress

- Trace the emergence of breeder blanket research, from early designs to current innovations.
- Evaluate the latest in breeder blanket technology and design and their use in different global fusion Research & Development (R&D) programs.
- Identify gaps in knowledge and highlight future trends and research priorities.

Review of Technologies and Designs

- Assess performance related to tritium breeding, thermal extraction and neutron shielding, with a focus on engineering trade-offs.
- Analyse and compare major breeder blanket concepts, including liquid metal and solid ceramic blanket designs.

Modelling and Multiphysics/Multiscale Processes

- Examine modelling techniques simulating neutronic, thermal and mechanical processes.
- Review multiscale and multiphysics tools for predicting and optimizing performance under fusion reactor conditions.
- Evaluate the present state of experimental techniques to validate simulations.

Referencing used in report is of Vancouver style.

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1. Introduction

Fusion energy promises sustainable and abundant energy, with breeder blankets playing a critical role in this pursuit. They serve as heat exchangers to produce electricity, neutron shields to protect reactor components and superconducting magnets, and tritium breeders to ensure fuel self-efficiency (1). The latter is crucial given tritium's limited global availability (2).

Early breeder blanket concepts, largely theoretical, focused on solid designs using materials like lithium and beryllium for neutron capture and multiplication. However, these faced challenges in heat extraction and structural stability (3). The International Thermonuclear Experimental Reactor (ITER) marked an important shift with its Test Blanket Module (TBM) program, testing breeder blanket concepts under fusion-relevant conditions. The program introduced concepts like the Helium Cooled Pebble Bed (HCPB) and Helium Cooled Lithium Lead (HCLL) (4). These early designs revealed material challenges, including tritium retention and helium production in ceramic breeders, which required further refinement.

Building on ITER's work, the EU Demonstration Fusion Power Plant (DEMO) is advancing four primary blanket concepts; Helium Cooled Pebble Bed (HCPB), demonstrating efficiency and a robust design (5); Water-Cooled Lithium-Lead Ceramic (WCLLC), employing a multiphysics modelling framework coupling computational fluid dynamics, heat transfer and solid mechanics (6); Helium Cooled Lithium Lead (HCLL); and Dual Coolant Lithium-Lead (DCLL). These concepts make use of advanced modelling, such as CAD-based parametric methods, to enable rapid iterations of 3D models for neutronics, thermal and mechanical analysis (7).

Global efforts extend beyond Europe, with the China Fusion Engineering Test Reactor (CFETR) tests innovative binary gas mixture cooling systems and tritium extraction (8). In the U.S., the ARIES program explores compact liquid metal systems, and tools like the Monte Carlo N-Particle (MCNP) code, influencing worldwide research (9). The SPARC project from Commonwealth Fusion Systems, in Massachusetts, focuses on compact, high-field reactor designs (10). In the UK, the STEP programme also develops breeder blanket concepts, with pre-conceptual designs expected to be finalized by 2024 (11), while the UKAEA's newly founded LIBRTI programme will be leading research through innovative testing methods (12).

Breeder blanket research is central to the success of fusion energy, with each concept addressing unique challenges in tritium breeding, heat extraction and neutron shielding. By integrating advanced modelling techniques and through international collaborations, breeder blankets are moving closer to operational readiness in commercial fusion reactors. This review evaluates state-of-the-art breeder blanket technologies, emphasising their role in making fusion energy a sustainable and economically viable reality.

Following this introduction, the report first provides insight into the fundamental functions and challenges of breeder blankets. It then provides an in-depth analysis of key solid and liquid breeder designs, noting the interplay between simulation tools and experimental programs.

2. Main functions of a Breeder Blanket

Tritium breeding ensures a continuous supply of fuel necessary for self-sustaining fusion. The deuterium-tritium (D-T) reaction is the most viable pathway for commercial fusion, yet tritium is exceptionally rare. Achieving a tritium breeding ratio (TBR) greater than 1, where tritium production exceeds consumption, is essential for operational viability. Blankets must effectively capture neutrons released from the fusion reaction and facilitate their interaction with lithium-based materials to generate tritium. Key challenges include neutron losses in blanket ports and tritium permeation through structural materials (13). Experimental systems like the ITER test blanket module aim to address these. The reactions responsible for tritium generation (14), the key behind self-sufficiency, are:

$$n + {}^{6}Li \rightarrow {}^{3}H + {}^{4}He + 4.79\ MeV$$

Eq. 1

$$n + {}^{7}Li + n \rightarrow {}^{3}H + {}^{4}He + n - 2.47\ MeV$$

Eq. 2

Research focuses on maximising TBR through neutron multipliers like beryllium or lead and improving extraction methods. The multiplication of neutrons with Be or Pb, responsible for maximizing the generation of tritium by increasing the number of neutrons and therefore the probability of reaction with lithium, are:

$$n + {}^{9}Be \rightarrow 2n + 2\,{}^{4}He$$

Eq. 3

$$n + {}^{208}Pb \rightarrow 2n + {}^{207}Pb$$

Eq. 4

The tritium cycle in fusion systems can be divided into two processes (15): the combustion cycle and the breeding cycle. The former involves D-T from the plasma being processed to remove unburned fuel and '*plasma ash*', then purifying and returning separated fuel to the fuel storage and delivery system. In the latter, fusion neutrons interact with lithium-containing breeding materials to produce tritium. It is then extracted using helium gas, purified in a recovery system, separated into hydrogen isotopes, and reintroduced into the system.

Breeder blankets also act as shields, protecting reactor components from intense neutron and gamma radiation. This means safeguarding structural materials, cooling systems, and the superconducting magnets which confine the plasma, from radiation induced damage and excessive heating. Neutron shielding reduces displacement damage and hydrogen/helium production in materials, extending their operational life (16). Common shielding strategies incorporate materials such as tungsten to attenuate gamma rays, and beryllium (17), or water, for neutron moderation.

The heat extraction role of breeder blankets involves the transfer of thermal energy to a power cycle, maintaining operational temperatures and ensuring structural integrity. Different coolants offer distinct advantages. Helium presents inertness and low neutron absorption; its efficiency is demonstrated in blanket designs like the ARIES-ACT2 (13). Dual coolant systems, such as the one present in DCLL designs, utilise Lead-Lithium (PbLi) for primary heat extraction and helium for First Wall (FW) cooling. As explained by Rapisarda, Fernandez (18), this configuration optimises thermal gradients and minimises the magnetohydrodynamic effects related to liquid metal coolants.

2. Neutronics, Modelling and Experiments

Fusion systems exhibit features including high neutron energy (14.06 MeV), intense flux levels and a complex, broad energy spectrum. Additional challenges arise from directional (anisotropic) scattering of neutrons, intricate material composition and large, geometrically complex reactor designs. With these features come challenges related to the neutronics analysis, warranting theoretical and experimental studies on neutron transport, neutron induced transmutation and activation, and neutron irradiation damage.

Neutron transport theory describes the movement and distribution of neutrons in a medium and is derived from the Boltzmann equation. It is used to predict neutron flux distributions (19), important for determining TBR and material activation. It is also a tool for optimisation of blanket geometry and material composition to enhance neutron capture. As with any

software, computing cost limits simulation capabilities. The transport equation is too complex to be solved analytically, requiring numerical approximations where the domain is split into small, discrete volumes that we can approximate. When discretizing, the number of variables greatly increase with the number of dimensions; this is known as the curse of dimensionality (20). All neutronics are based on nuclear data libraries, the foundation of research and design of nuclear systems (21). They are comprehensive archives containing information about the interactions of neutrons and other particles with various materials. For example, cross sections, the measure of likelihood of a specific interaction between a particle like a neutron and a target nucleus.

Key assumptions are made deriving the neutron transport equations. Neutrons are considered as point particles with no charge, travelling in straight lines between collisions. Neutron-neutron interactions are deemed negligible, and collisions are taken as instantaneous. Material properties are assumed isotropic and the nuclear properties and compositions of materials under consideration are taken as constant. Finally, only the expected/mean value of neutron density distribution is considered. To describe the model, you need to define the neutron distribution in space, energy, direction and time. The result is shown in Eq. 5, the neutron transport equation, and more complex explanations of derivations and complete forms of the equation are detailed by Sanchez, McCormick (22):

$$\frac{\partial n}{\partial t} = Production\ rate(Q) - Leakage\ rate(L) - Removal\ rate(R)$$

Eq. 5: Neutron Transport Equation

The equation can be solved as either a fixed source problem or an eigenvalue problem, depending on the presence of an external neutron source. In fixed source problems, neutrons are supplied externally, and over time, the neutron population stabilises as production balances losses due to absorption and leakage. This is always the case for fusion. In eigenvalue problems, there is no external source, and neutrons are generated solely through internal reactions. The system's behaviour is then classified as subcritical, critical, or supercritical, based on the effective multiplication factor (k_{eff}) as subcritical (k_{eff} <1), critical (k_{eff} =1), or supercritical (k_{eff} >1). These correspond to scenarios where the neutron population decreases over time, remains steady, or increases exponentially, respectively. (20).

Numerical methods work around the difficulty of analytical solutions, approximating the transport equation. Monte Carlo (MC) methods, which use stochastic sampling, are particularly effective in handling the complex geometries and anisotropic scattering

characteristic of fusion systems. Their advantages lie in their use of continuous-energy cross-section and accurate geometric descriptions. These methods suffer from slow convergence rates, particularly in deep-penetration scenarios, which may limit their computational efficiency for large-scale fusion systems. The underlying principle involves simulating numerous neutron interactions, recording outcomes, and using statistical averages to estimate probabilities or system behaviours (21).

Examples of MC transport simulation codes include the The Super Monte Carlo Program for Nuclear and Radiation Simulation *SuperMC* and the Monte Carlo N-Particle transport code *MCNP*, both used in different initiatives worldwide as is detailed in further sections. SuperMC was developed by the FDS team of the Institute of Nuclear Energy Safety Technology in Hefei, China. As explained by Wu, Song (23), it can simulate multiple types of particles (neutrons, photons, electrons and protons), considering the thermal scattering of neutrons. It has seen widespread use in evaluating nuclear heat deposition in the Toroidal Field Coils (TFCs) of ITER, for example. It has also been implemented for nuclear analysis and optimisation of shielding of observation systems in vacuum vessels, assessment of radioactive waste, and tritium breeding rate of blankets.
The MCNP code was developed in Los Alamos National Laboratory. As detailed by Zieb, Hughes (24), it can be used to perform transport computations of heavy charged particles. It can perform eigenvalue and fixed source calculations and has been used for neutronic analysis at ITER; neutron wall loading, neutron/photon fluence rate and nuclear heat deposition have all been modelled using MCNP, as well as TBR calculations (25). It sees extensive use by the UK Atomic Energy Association (UKAEA) for analyses involving their Spherical Tokamak for Energy Production (STEP) prototype (26).

Deterministic methods solve the neutron transport equation by discretising variables into mathematical-physical equations, offering faster computations compared to MC methods. Their application is constrained by limited flexibility with irregular geometries and the anisotropy of neutron scattering. To overcome these challenges, hybrid approaches, such as the Consistent Adjoint Driven Importance Sampling (CADIS) method, combine the accuracy of MC simulations with the efficiency of deterministic solvers. This approach has proven effective for ITER-scale neutronic analyses, demonstrating its capability to balance computational speed and geometric complexity (27). Despite this success, further validation would be necessary to assess the scalability for systems like DEMO, with higher fluxes and more intricate geometries.

Neutronic models are inherently imperfect and rely on experiments for validation. Early tests focused on fission systems, so the introduction of fusion systems meant new requirements for experimental data. The D-T reaction produces neutrons of approximately 7 times the energy of fission neutrons (at 14.06 MeV). For fusion the focus is on basic research; material activation, tritium breeding and material irradiation damage and shielding investigations. These are necessary to validate accuracy of computational methods, the precision of nuclear data libraries and reliability of neutronics design choices. Experiments can be broken down into different types:

- Material irradiation damage and shielding experiments specialise in simulating irradiation conditions of mixed fields of neutrons and photons. ITER, for example, is designed to produce a neutron yield of an order of around 10^{20} per second (28), which affects components like the FW, blankets and superconducting magnets, requiring chosen materials (29) to undergo testing. The magnets confining the plasma are especially sensitive to neutron irradiation.
- Material activation experiments are necessary to acquire certain parameters such as neutron reaction channels, reaction cross sections and half-lives of the nuclides (30).
- Tritium breeding experiments are necessary to validate the blanket neutronics satisfies requirements. Since self-sufficiency is essential for steady state operation of fusion systems, a scientific design combining breeding and extraction is required. The procedure involves a blanket experiment module being made, either full-scale or scaled down. Several equal spacing measurement channels are oriented with the incident direction of neutron beam. Samples are placed into measurement channels and are irradiated (31). Since the tritium production rate is related to the neutron spectrum (the former can be calculated from the latter) the measurements can be mutually validated.

Experimental facilities are currently limited by the neutron sources available. Point neutron sources, with fusion-relevant energies (but lower flux), and high-flux neutron sources are difficult to integrate into one. Most current breeding experiments are still based on “point” neutron sources of low intensities, with measurements of tritium production rate being able to be performed when the experiment is done. A large gap exists between these small scale rigs and a realistic fusion environment, and further verification of real-time tritium extraction during the experiment is needed. In the past, mock-ups of the helium-cooled pebble bed and helium-cooled lithium lead have been irradiated at relatively low neutron flux of approximately $5e10^{8}$ $n/cm^2/s$, as compared to ITER’s levels, which are anticipated to reach $10^{14} n/cm^2/s$ (32). It is therefore necessary to develop “volumetric” neutron source

experimental facilities to conduct online tritium breeding and extraction experiments, verifying the effectiveness of the design of the tritium breeding modules.

Initiatives worldwide aim to solve these issues. LIBRTI is a new flagship programme which involves building a new facility and perform experimental research (12). Its aim is to research how to optimise TBR for a range of blanket concepts and how to achieve this in a reproducible and predictable way. Not limited to this, it will be designed to become a data acquisition centre: the objective is to create a test site where as many experiments as possible can be undertaken. There are severe gaps in our knowledge and the only way to mitigate this is to do as many experiments as possible. They will be obtaining a high-reliability neutron source of as high flux as manageable. It will be an international consortium collaboration, with an initial focus on areas such as liquid metal, molten salts and solid blanket designs.

The International Fusion Materials Irradiation Facility – Demo-oriented Neutron Source (IFMIF-DONES) represents a crucial initiative for experimental validation of simulations of irradiation, displacement damage and material activation. Bernardi, Ibarra (33) describes how it mimics the neutron spectrum, with energies up to 40 MeV, and flux conditions anticipated in the FW of DEMO reactors. It employs a high-intensity deuteron beam on a liquid lithium target and generates neutron fluxes capable of reproducing nuclear responses seen in fusion environments, a step in the right direction of closing the divide between theoretical models and operational reality. It addresses the long-standing lack of volumetric neutron sources. It is worth noting that such high energy neutrons would not be representative of a fusion reactor environment. For fusion-relevant conditions, attenuation is necessary: the peak will be at 14 MeV and, although the flux will be large enough, there will be a high energy tail above the peak.

3. Solid Breeder Blankets

Solid breeder blankets utilise ceramic materials like lithium orthosilicate (Li_4SiO_4) and lithium methatitanate (Li_2TiO_3) to breed tritium through neutron interactions. These ceramics are typically configured into pebble beds to maximise neutron capture efficiency, and beryllium is often incorporated as a neutron multiplier to enhance TBR (Pupeschi, 2017). Prominent concepts include the Helium Cooled Pebble Bed (HCPB) and the Water-Cooled Lithium Lead Ceramic Breeder (WCLCB), extensively developed in the EU Demo. China and South Korea contribute to unique designs like the Helium-Cooled Ceramic Breeder (HCCB) and Helium-Cooled Ceramic Reflector (HCCR) respectively.

3.1 Helium Cooled Pebble Bed

The HCPB stands out as one of the key solid breeder blanket concepts proposed for fusion reactors like the European DEMO, which leads the progress in research. Its design involves the use of lithium-based ceramics as tritium breeders and beryllides as neutron multipliers.

Initial progress is marked by the ITER Test Blanket Module (TBM), featuring a robust grid structure forming cubicles for breeder and multiplier materials: a ‘beer-box’ design. It was structurally sound, yet large amounts of steel in the Breeder Zone (BZ) compromised its tritium breeding performance (5). Tailored for DEMO’s requirements, a less complex ‘sandwich’ design arose after a pre-conceptual (PCD) iterative design phase (34). It had alternating layers of tritium breeder and neutron multiplier, separated by cooling plates made of Eurofer97 steel. However, complexity of manufacturing regarding small channels embedded in cooling plates and the large pressure drop of the channels resulted in an improved fuel-breeder pin architecture being proposed (35).

The novel design involves cylindrical ‘pins’ that house the tritium-breeder material in a hexagonal matrix. These breeder pins are embedded in prismatic blocks of beryllides, themselves surrounded by Eurofer97 structural steel, providing mechanical strength and acting as a thermal conduit (36). Figure 1 illustrates the evolution from the complex ‘beer-box’ design to the ’sandwich’ and fuel-breeder pin configurations.

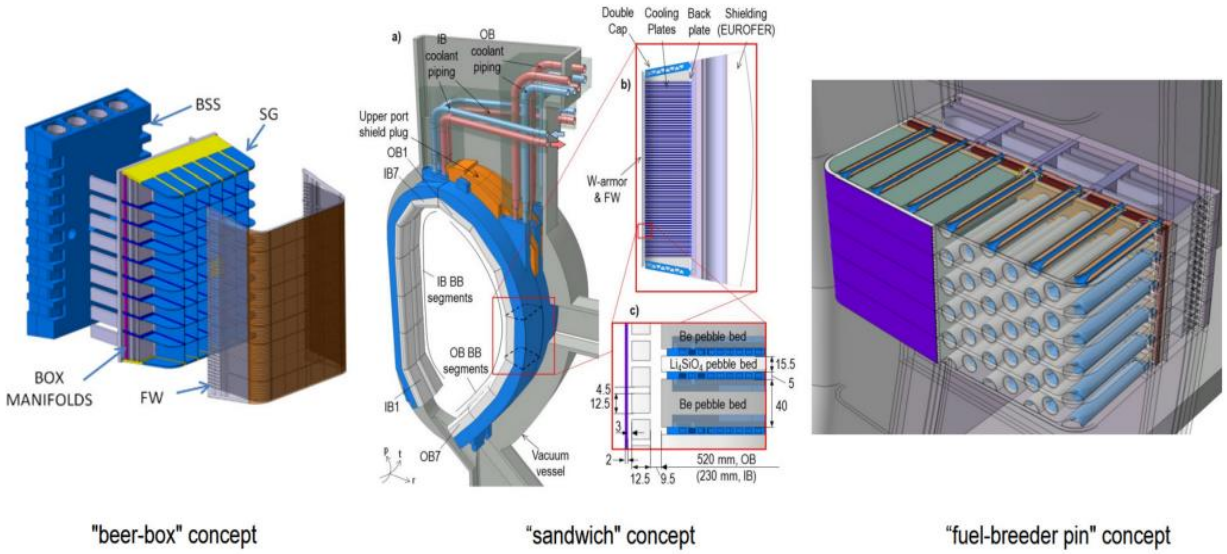


Figure 1: Design evolution of HCPB BB During PCD Phase (5)

A pebble bed is used, composed of a lithium silicate and lithium titanate mixture (Li_4SiO_4 + 35 % mol Li_2TiO_3), also known as the Advanced Ceramic Breeder (ACB). These ceramics are selected for their favourable thermos-mechanical properties, high lithium density, and

chemical stability under high-temperature conditions (37). The ceramic breeder's ^{6}Li is enriched at 60% to optimise tritium breeding performance, responsible for the primary tritium breeding reaction (displayed in Eq. 1). For the Neutron Multiplier Material (NMM), the beryllide $Be_{12}Ti$ is used due to its positive neutronic characteristics and enhanced safety (38). It serves dual roles as a neutron multiplier and a first layer of shielding, scattering and slowing down high-energy neutrons and improving the shielding. Nevertheless, current designs only marginally meet attenuation requirements. As highlighted by Hernández, Pereslavtsev (35), there is a need for more compact and efficient shielding materials.

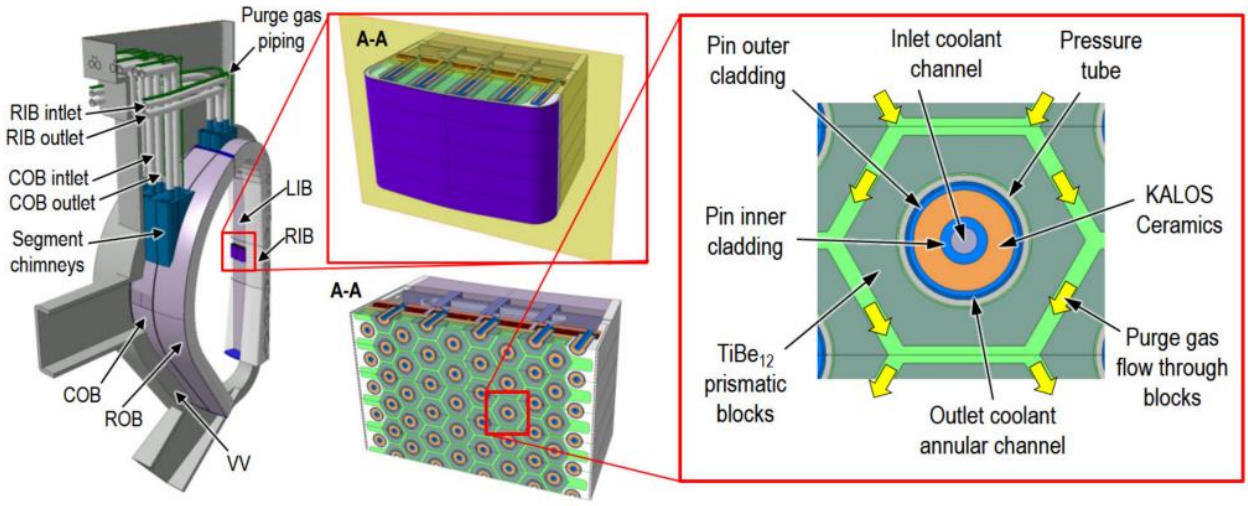


Figure 2: HCPB Breeder Blanket Schematic at end of PCD Phase (5)

The fuel-breeder pins consist of concentric inner and outer claddings, separated by an annular space filled with a pebble bed of the advanced ceramic breeders. These are investigated at the Karlsruhe Lithium Orthosilicate (KALOS) Facility (39). Surrounding the pins there is a layer of beryllide material. This arrangement optimises tritium breeding while enhancing neutron economy. The inner cladding features longitudinal grooves to channel the helium purge gas, facilitating tritium and heat extraction (35). Figure 2 aids visually understanding the configuration of the pins, with a cross-section view.

Heat is conducted through the ceramic layer and structural walls to the helium coolant. The pin claddings and surrounding structural frameworks are made of Eurofer97 steel. This ferritic-martensitic alloy is chosen for its combination of high thermal conductivity, low activation properties and exceptional resistance to radiation-induced swelling and embrittlement. However, Hernández, Pereslavtsev (34) raises concerns; at high temperatures with trace impurities in the helium coolant it is prone to oxidation. There is a

notable lack of experimentally validated mitigation strategies, which should be addressed soon.

Aluminium-based and ceramics protective coatings have been proposed to mitigate oxidation risks and to extend the operational life of Eurofer97, with Spain's *Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas* (CIEMAT) carrying out relevant experiments as explained by Hernández, Sánchez (40). Experimental studies on helium-water mixtures to improve tritium extraction efficiency while minimizing permeation, following the findings of Hernández, Pereslavtsev (38), constitute another avenue to tackle those Eurofer97 oxidation issues.

Heat generated is conducted through the pebble beds and walls into the helium coolant. The integrated cooling plates and helium manifold system ensure a uniform and efficient heat removal process. The heat extraction process is optimised through a combination of advanced thermohydraulic features and innovative structural designs. The FW is the plasma-facing component of the blanket, responsible for absorbing heat and radiation from the plasma. To enhance its heat transfer, it incorporates V-ribs and surface roughening as turbulence promoters, increasing heat transfer coefficients (HTCs) while maintaining low pressure drops ($\Delta p_{FW} \approx 1\ bar$) (35). Computational analyses use complex computational fluid dynamics (CFD) for large eddy simulation and scale adaptive simulation to accurately represent the flow physics, however they are very computationally costly and cannot represent a full-scale FW model (8). They indicate that the implemented features can handle FW heat loads of up to 1.4 MW/m^2, ensuring material temperatures remain below 550ºC (35).

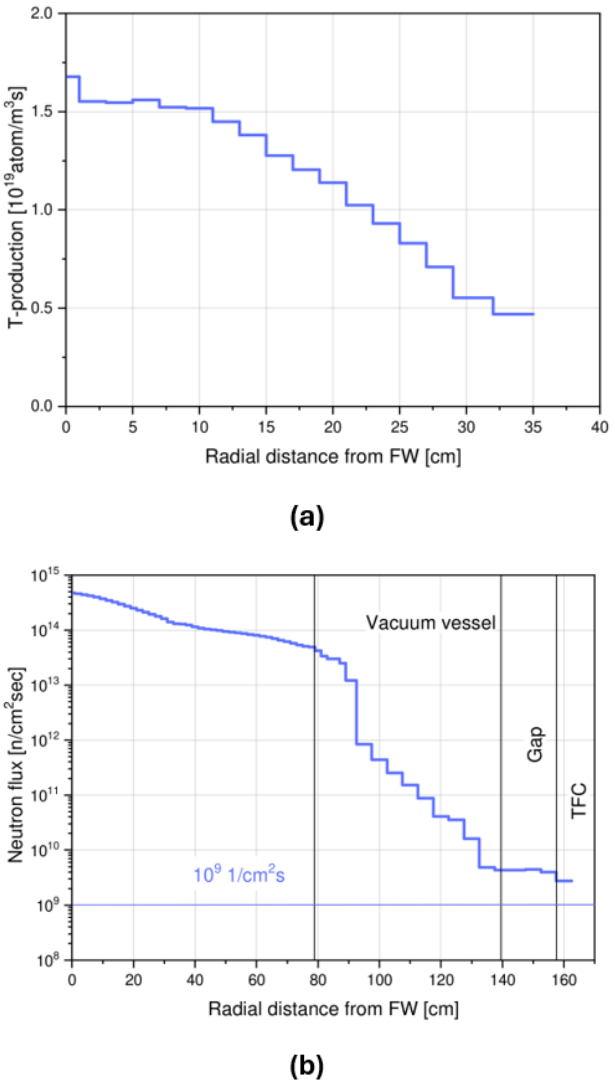


Figure 3: (a) Tritium generation rate profile; (b) Radial neutron flux profile (5)

The coolant, helium gas, presents excellent thermal properties, compatibility with structural materials, and lack of activation under neutron irradiation (13). The temperature difference (300 – 500ºC) allows for efficient energy transfer to a secondary power generation circuit. It moves at low flow speeds (<10

m/s) through the annular space of fuel-breeder pins, demanding additional roughening of the cladding surfaces to maintain reasonable HTCs despite the reduced Reynolds number (35). CFD simulations and empirical models, such as the Bhatti-Shah correlation for roughened channels, validate the results. Main challenges include local hotspots near the FW due to sudden helium expansions. This effect is not accounted for in the simplified thermal model and is therefore a matter requiring future investigation.

Studies for alternative designs for CFETR blankets have explored He-CO_2 mixtures as coolants to address helium's lower density and high drive power requirements. Guo, Wu (8) propose a 60% He, 40% CO_2 which reduces drive power by nearly 89.36% while maintaining appropriate thermal properties in tritium breeding zones. Despite promising results, the chemical reactivity of CO_2 at high temperatures limits its operating temperature to under 750ºC, potentially restricting performance in high-temperature applications.

For analyses concerning tritium breeding ratio, nuclear heating, shielding efficiency, activation and irradiation damage, an MC code (MCNP5-1.60) was used. A CAD-to-MC conversion tool (McCad) is used to obtain 3D heterogeneous models to be analysed, with high fidelity of the CAD design. Although Hernández, Pereslavtsev (35) lays out the high-resolution insights these provide, their computational cost limits full-scale modelling. Zhou, A. Hernández (5) emphasise the importance of experimental validation to reduce reliance on simplified thermal and neutronics models.

A tritium breeding ratio (TBR) of 1.20 is calculated as the achievable value, satisfying the TBR requirement of 1.15 comfortably, allowing for inherent uncertainties in modelling and nuclear data, as well as coverage loss due to penetrations (5), (38). Numerical models can be used to display tritium generation rate as a function of radial distance in the blanket, as shown in Figure 3a. This data is then used as input for tritium transport analyses (5). Results include the radial neutron flux (Figure 3b), which demonstrates the vacuum vessel (positioned beyond the blanket) is one of the main neutron-shielding components for the toroidal field coils (TFC) which generate the magnetic fields confining the plasma.

The current design facilitates efficient tritium extraction; Helium, also used as the coolant, serves as a carrier gas to sweep the generated tritium from the blanket for recovery. However, another significant challenge involves the tritium permeation through Eurofer97 into the helium coolant. While Hernandez, Arbeiter (41) suggest doping helium with 0.1% hydrogen or exploring water as an additive, the trade-offs between improved recovery efficiency and increased system complexity need further investigation.

3.2 Water Cooled Ceramic Breeders

Water-Cooled Ceramic Breeder (WCCB) blankets are a promising driver candidate for fusion reactors. By utilising water, a well-established coolant in conventional fission reactors, WCCBs manage the significant thermal loads imposed by fusion conditions while simultaneously enabling tritium breeding and neutron shielding.

The relevance of WCCBs extends across major fusion programs, with tailored designs reflecting regional goals and research priorities. In China, the CFETR incorporates compact coolant manifolds and optimised BZs to enhance tritium breeding and thermal performance (42). In Europe, it is actively being developed as an alternative to water-cooled lithium-lead blankets as part of its DEMO strategy (43). Meanwhile, Japan's JA DEMO program uses mock-up tests under fusion-relevant conditions through the ITER TBM programme, bridging conceptual designs with operational readiness (44).

Development of WCCB blankets has been a progressive journey. Early research into WCCB, like that of the HCPB, was done through the ITER TBM program, providing a platform to test under fusion-relevant conditions. Japan's WCCB-TBM conceptual design, approved in 2016 (45), initiated a series of structural and functional design reviews aimed at meeting both ITER and DEMO requirements. Initial designs adopted box-shaped modules. Thicker walls were necessary to keep the integrity of TBM in case of an in-box Loss of Coolant Accident (LOCA), which is when one or more cooling pipes in the BZ rupture, and high-pressure coolant flows into the BZ and further along with the purge gas, endangering the blanket box and the tritium extraction system. (46). However, thicker walls would also mean reduced tritium breeding capability since the tritium-breeding material volume would decrease. Moreover, the weight limitation of the Reduced Activation Ferritic Martensitic (RAFM) steel, named F82H, was already being exceeded. These issues led to a cylindrical submodule configuration to be devised. This novel design, as explained by Kawamura, Gwon (45), uses 56 submodules, each with a cylindrical container. They involve pebble beds containing lithium ceramic (Li_2TiO_3) for tritium breeding and Beryllium as the neutron multiplier. Pressurised water at 15.5 MPa flows through built-in cooling channels within the hemispherical walls of the submodule. One key aspect of WCCBs is how they leverage the

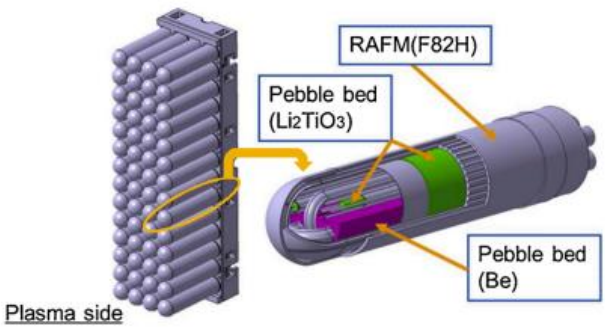


Figure 4: Novel Cylindrical Submodule design (45)

well-established principles of pressurized water reactor (PWR) technology (42), with water operating at temperature ranging from 285 to 325 ºC in the Japanese design. Cooling pipes are embedded within the pebble beds for efficient heat extraction, ensuring the module withstands both surface heat flux and neutron wall load, of 0.3 and 0.78 MW/m^2 respectively. Figure 4 includes a schematic of the latest design.

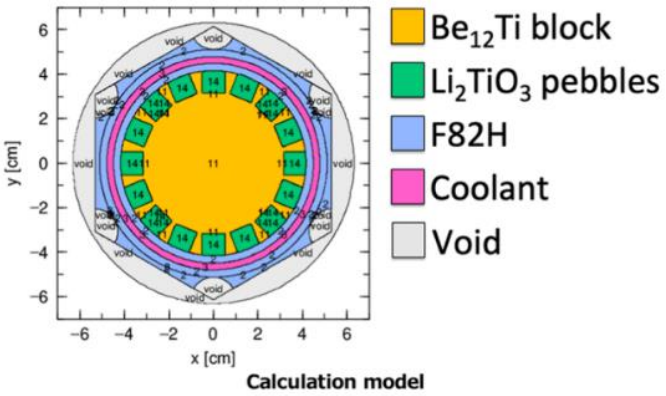


Figure 5: Latest JA DEMO design cross section view for calculation model (44)

Table 1 highlights the material optimization achieved in the cylindrical design, reducing F82H usage by 60%. This not only meets ITER's vacuum vessel constraints but also increases the volume available for breeding materials, enhancing TBR performance.

Finally, to simplify the structure of their design, the neutron multiplier (Beryllium) was changed from a pebble bed form to a block shape, enhancing heat transfer. The resulting final design is pictured in Figure 5. The local TBR was therefore calculated to have a value of 1.21, exceeding the target value of 1.19 (44). An 'envelope' concept, presented in the same paper, involves adding a thin layer of tritium-breeding material around the structural steel. This is meant to minimise the neutron absorption by the steel. However, precisely laying a ~3 mm layer of ceramic around steel components involves complex manufacturing. Additive manufacturing, although challenging due to high melting temperatures and inherent brittleness, could be explored as an option (47). Another option is physical vapor deposition, which is already used in industries such as aerospace to produce thin, uniform ceramic coatings (48). These envelopes will require thermal-neutronics simulations to analyse stress distribution under high thermal gradients, as well as investigating thermal expansion compatibility between ceramics and steel to minimise delamination risks.

Table 1: Material Optimization in Redesign (45)

Materials	Box-shape	Cylindrical	Ratio
F82H	∽1950 kg	791 kg	0.41
Li2TiO3	48 kg	61 kg	1.27
Be	230 kg	129 kg	0.56
Water	38 L	9 L	0.24

MCNP and SuperMC codes are coupled with thermal-hydraulic and structural models provides a more comprehensive understanding; Li, Li (49) describe CFETR's use of a model (50) which Qu, Cao (51) used to optimise TBR, finding an increased value from 1.219 to 1.282. They were able to precisely adjust the BZ

configurations, incorporating Li_2TiO_3 as breeder material and $Be_{12}Ti$ as neutron multiplier. The latter provides higher neutron multiplication and thermal stability compared to pure Beryllium. Lin, Cheng (46) conduct a study using the RELAP5 code (used in transient thermal-hydraulic analysis in pressurized water reactors) to analyse the performance of a single WCCB module: they discuss how coolant channel geometry impacts pressure drops and temperature distribution. Narrower channels increase volume available for breeding material, key in the increase observed, but also add hydraulic resistance, leading to higher pressure drops and potential hot spots.

Analysing the TBR observed in the DEMO report by (43), we find slightly lower values than those in Japan and China. They obtain a TBR of 1.17, which still meets the 1.15 requirement set in their program. We can find several reasons which may explain the lower value. First, their breeder material is a pebble bed of Li_4SiO_4 + 35 % mol Li_2TiO_3. This contrasts with Japanese and Chinese designs which use pure or higher fractions of one material for the BZ. Lithium orthosilicate, making up most of the DEMO's pebble bed, has a higher lithium density compared to lithium titanate, with superior tritium release at lower temperatures, yet has lower thermal conductivity and reduced thermo-mechanical stability (52). Europe's mixed approach balances each material's properties, sacrificing some tritium production for mechanical performance.

Moreover, the integration of the Eurofer97 structural steel (43), enhancing reliability under neutron irradiation, impacts neutron absorption and takes space from tritium breeding material, reducing TBR. Safety is a cornerstone of the DEMO program, leading to conservative designs with built-in redundancies. This may result in the DEMO program falling short in some respects as compared to other initiatives, forced to meet low activation and structural reliability goals. An example is CFETR's compact manifold design, highlighting its focus on maximising TBR at the cost of increased engineering challenges related to operational reliability and manufacturing complexity. The difference in philosophy in DEMO is reflected by their research on shielding design optimisations, investigating neutron absorbers such as boron carbide (B_4C). From the baseline to implementation of the idea presented by Palermo, Hernández (53), TBR progressively suffers a decrease. This loss, although marginal in practice (0.26%), emphasises their priorities in maximising safety.

However, this balance represents a nuanced trade-off, and there are benefits to DEMO's strategy. Their WCCB design preserves many of the high performances of the original HCPB concept. Zhou, Lu (43) state the direction of their research will be oriented towards

assessing simpler architectures, taking advantage of the flexibility that a liquid neutron multiplier provides. The same paper mentions the possibility of using lead-alloys (such as $LaPb_3$) as an alternative neutron multiplier. From recent studies into this material (54), lanthanum plumbide ($LaPb_3$) would require extensive investigation for its integration due to its rapid degradation due to corrosion, which could mean having to include more steel into the design to protect it. This could be detrimental to the volume available for tritium breeding material.

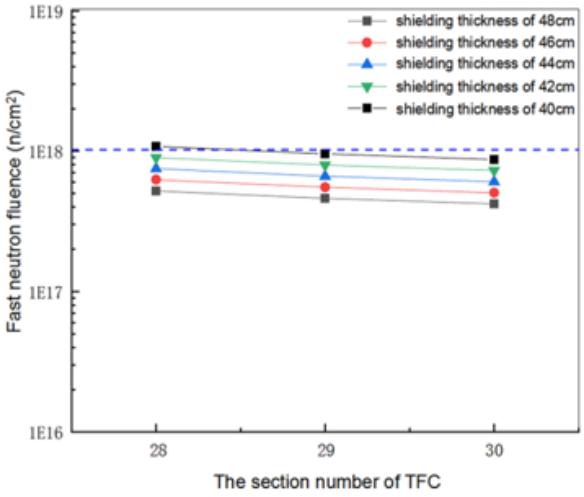


Figure 6: Fast neutron fluence in a coil segment of the TFC in insulator (49)

In contrast, CFETR prioritises minimizing shielding material to reclaim space for the breeding zone. Through a fast neutron fluence distribution analysis, Li, Li (49) were able to optimize modules to decrease the shielding thickness. An initial value of thickness of around 48 cm saw reductions of up to 6 cm in some modules, reallocating this volume to the BZ. Figure 6 shows one of the plots for fast neutron fluence (the integral of neutron flux over time – neutrons/cm^2) for shielding thickness values in the insulator and layer of toroidal field coils (the most susceptible part of the reactor to irradiation damage). The blue dashed line represents the maximum allowable neutron fluence limit as set by ITER; it is important to note that the design never exceeds the limits considered safe by Europe's initiative.

Heat extraction in WCCB designs is a critical factor in maintaining operational stability and tritium release efficiency. DEMO adopts a conservative approach, with well-distributed coolant channels operating at 15.5 MPa and 285–325°C (like Japan's design), ensuring uniform heat removal but limiting BZ volume (43). In contrast, CFETR employs compact manifolds that maximise breeding material volume but introduce challenges such as increased hydraulic resistance and localised overheating (46). RELAP5 simulations have been pivotal in validating these designs, ensuring coolant flow remains stable under high-pressure drops. Japan's cylindrical submodules, tested in ITER TBMs, integrate built-in cooling channels to optimise thermal performance while accommodating thermal and structural loads (44). These approaches highlight the trade-offs between stability, efficiency, and tritium breeding across WCCB designs.

4. Liquid Breeder Blankets

Liquid breeders rely on a lead-lithium (PbLi) eutectic alloy as a coolant and, in some designs, as the tritium breeding medium (18). They offer high breeding efficiency and simpler tritium extraction than their solid counterparts. The integration of a liquid metal environment is challenging, with unique challenges such as magnetohydrodynamic (MHD) effects under strong magnetic fields and material compatibility issues due to corrosion and erosion (55). The Helium-Cooled Lithium-Lead (HCLL) concept employs a helium coolant loop to manage thermal loads, while circulating PbLi in dedicated channels ensures tritium generation. In contrast, the Dual-Coolant Lead-Lithium (DCLL) design, which this section will focus on, uses both helium and innovative PbLi cooling circuits for thermal management, neutronics and performance, addressing limitations of single-coolant designs. It has seen significant development in the US (56).

4.1 Dual-Coolant Lithium-Lead (DCLL)

The DCLL breeding blanket employs a PbLi eutectic alloy for tritium generation and heat removal, augmented by a helium coolant loop to control structural temperatures. Iterative design improvements, such as flow channel inserts and advanced materials, seek to balance efficient tritium breeding, thermal performance and effective neutron shielding.

Over the past two decades, the DCLL blanket has evolved through design iterations and operational strategies across Europe, the US and China. Low-temperature (LT) concepts employ the eutectic PbLi alloy cooled below 550 ºC and use sandwich-type Flow Channel Inserts (FCIs), which act as electrical and thermal insulators, with alumina layers surrounded by Eurofer steel. They are limited by the maximum allowable operating temperature of structural materials (18). High-temperature (HT) concepts, with fluid at around 700 ºC, incorporate SiC-based FCIs to provide both electrical and thermal insulation. They enable higher thermodynamic efficiencies (around 45% vs. 34% for LTs) as they are compatible with Brayton cycles (55). However, these high-performance configurations demand significant R&D into the irradiation behaviour of SiC, the fabrication of complex FCIs and the resolution of corrosion and erosion challenges. Under the EUROfusion program, CIEMAT has led integrated engineering efforts (57), including defining system specifications, conducting CAD modelling and performing neutronic, thermal-hydraulic and MHD analyses.

The PbLi flows at velocities ranging from 0.1 – 1 m/s (18) through the breeding zone at inlet/outlet temperatures near 300/550 ºC, acting as the primary coolant and tritium

breeder. Helium, introduced separately at inlet/outlet temperatures of 300/420 ºC, provides focused cooling for the FW and other structural components. Optimised helium flow velocities, on the order of tens of meters per second, support high heat transfer coefficients (up to 6000 W/m^{2}K), enabling the FW to stay within operational temperature limits (58).

To mitigate the MHD pressure drops induced by the magnetic fields (of around 5 T), FCIs are employed, limiting electrical conductivity. In LT concepts, an alumina layer is surrounded by RAFM steel (Eurofer). These design choices result in a calculated TBR value of around 1.13 (18), and energy multiplication factors (M_E) near 1.195, confirming their capability to meet fuel self-efficiency and energy conversion requirements. HT variants utilise a 5 mm thick SiC-based design which, as explained by Smolentsev, Morley (55), are separated from the RAFM steel channel walls by a 2 mm channel gap filled with PbLi. They provide electrical and thermal insulation, allowing higher outlet temperatures (around 700 ºC) and improved thermal efficiencies, compatible with advanced energy conversion systems like Brayton cycles. However, they require continued R&D in SiC irradiation tolerance, FCI fabrication technologies and corrosion mitigation strategies (55). Much of the early MHD analysis relied on simplified 2D assumptions; current modelling increasingly addresses 3D phenomena and accounts for turbulence, non-uniform fields and coupled flow–heat-transfer behaviour (56). Outlined by Rapisarda, Fernandez (18), segmenting the BZ into four parallel PbLi circuits ensures more uniform neutron flux distribution, stable flow patterns and minimized reverse flow regions. It facilitates reliable neutronics performance, with PbLi extracting around 60% of the blanket's power and helium handling the remaining 40%.

This concept also covers the role of shielding, providing neutron attenuation. Dense structural regions containing Eurofer, helium and PbLi within Europe's Back Supporting structure (BSS) explained by Rapisarda, Fernandez (18) enhance shielding, ensuring neutron heating limits in susceptible superconducting coils are not surpassed. Preliminary neutronic analyses from the same report show that neutron wall loadings of around 1 MW/m^2 can be achieved while maintaining TBR of over 1.1, fulfilling self-sufficiency criteria. However, operational conditions involve sustained irradiation doses (assumed to be of 20 dpa in steels over 6.6 years) and intense thermal, mechanical and chemical stresses. RAFM steels and SiC-based FCIs in other designs, as presented by Smolentsev, Morley (55), face corrosion, erosion and swelling challenges, magnified by relatively high PbLi flow velocity, complex temperature gradients and the presence of magnetic fields.

Recent efforts outlined in the same report introduced corrosion limits (e.g., less than 20 μm/yr wall thinning) and foam-based SiC FCIs to mitigate mechanical and irradiation-induced stresses (59), but studies are still needed to form representative models in fusion-relevant conditions.

State of the art computational tools are being developed and refined to tackle the DCLL's multiphysics challenges. Neutronics simulations, for example using MCNP (60), coupled with CFD and MHD codes, such as HMAG (56), enable the assessment of complex flow regimes, tritium transport and corrosion processes. Sensitivity studies using MATLAB Simulink (18) examine the impact of inlet temperatures, flow rates and channel geometry on global performance metrics. Moreover, integrated workflows that incorporate volumetric heating data, advanced meshing and parallel computation accelerate design iteration and optimization (58).

Simulations are gradually being anchored in experimental data. The MaPLE loop at UCLA (61) replicate MHD flows, thermal conditions and magnetic fields to reduced scale, providing critical validation points for MHD models and revealing discrepancies in simplified assumptions. Meanwhile, Favuzza, Aiello (62) explain how the Lifus 6 plant in the Brasimone Centre in Italy tested RAFM steels, like Eurofer97 and F82H, for their erosion-corrosion resistance in contact with liquid lithium. The tests simulated reactor-relevant conditions with varying levels of impurities, flows of 15 m/s and temperatures of 330 ºC, exposing the materials for different lengths of time. The materials exhibited great resistance with corrosion rates under 0.2 μm/year, significantly under the IFMIF target of 1 μm/year. The findings serve as a benchmark for validating erosion-corrosion models experimentally to ensure long-term component durability.

As the DCLL concept matures towards DEMO applications, key priorities include advancing SiC FCI fabrication techniques, validating irradiation induced phenomena, such as swelling and creep, and characterizing corrosion under high velocities and strong magnetic fields (55). The same studies suggest that further tritium extraction studies, permeation barrier development and refined Multiphysics models will be critical for closing performance gaps.

5. Review of the use of generative AI

Generative AI, specifically ChatGPT 4o, was used to facilitate research and aid in the understanding and structuring process of this report, as well as to help in structuring the report and polishing the language.

During the initial research phase of my LRP, I input my objectives and the general guidelines for the project into ChatGPT. I then fed it papers I was finding and asked it to provide me with the key takeaways that it deemed relevant to my LRP. Although promising at first, I realized it functions with an ‘alignment bias’: which is to say it will always look to justify the premise it is presented with. For example, if asked to explain how a certain software is used in a field, it may fabricate an example that is not present in the sources it is supposedly referencing. Its bias may appear obvious in some answers, but more dangerously it could be a result of a longer conversation, where a prompt entered days or weeks before, long forgotten by the user, affects the answers the AI is providing. Once my research had progressed, and I was ready to start writing down my ideas, I turned to AI for help again. ChatGPT 4o allows you to create custom ‘GPTs’, to which you can feed information and set guidelines, a specific set of tasks. By introducing the literature I had found up to that point, as well as the rubric of the LRP, I created an ‘expert’ on the topic I had to cover. I then asked for it to lay out a structure that would cover the main objectives I had set out with my supervisor. While I had to change it significantly, it helped me start writing down my ideas, as it is often easier to begin when you have an initial structure.

While writing my main discussion, I found further use of ChatGPT in finding sources from the web in reputable scientific journals (which I had to specify explicitly) for different designs or concepts I otherwise would have had to look up myself. Even when the papers supplied were not exactly related to my requests, it was still very useful to explore new literature and discover important references in the field from the articles it did provide. The AI was also of use in polishing the language. It helped me understand when I was transitioning smoothly, or not, between topics, and where I was losing focus or lacking conciseness. Although its capability of creating content from the base research was lacklustre, when presented with content I had already written it was able to improve the style and enhance clarity.

In conclusion, when aware of the clear limitations generative AIs have, I found that it is possible to work around some of them, such as the confirmation bias, making them a very useful aid in writing a literature review.

6. Conclusion

Breeder blankets are key in advancing fusion energy as a viable power source. This report has reviewed their development, spanning from early solid breeder concepts to advanced liquid metal designs. It focuses on key parameters in evaluating operational readiness of

different configurations, such as tritium breeding, heat extraction and neutron shielding capabilities. The work of initiatives around the world highlights the global effort to refine the technology, in collaboration with experiment-focused programmes which push toward fusion-relevant experimental validation.

Achieving TBRs above unity, mitigating radiation-induced damages and minimizing design-specific concerns such as corrosion, erosion and MHD-induced effects remain primary engineering challenges. Advanced neutronics simulations with the use of software like SuperMC or MCNP have improved predictive capabilities, with multiphysics and multiscale frameworks streamlining iterative design processes. Experimental facilities like the IFMIF-DONES provide critical validation for models. They guide material selection and substantiate nuclear data libraries' information.

Across the examined designs, TBR has reached calculated values of up to 1.282, in the CFETR's WCCB configuration. Thermal management has also been optimised, exemplified by innovative dual-coolant systems in the DCLL blanket, which is designed to operate at temperatures ranging from 550 to 700 ºC. Shielding is integrated with structural components, such as the back supporting structures of the DCLL design.

Although we can use the neutronic analyses results as guidance to evaluate blanket concepts, it is difficult to ensure assumptions are the same across different designs in different programmes. This means comparisons are difficult. However, the HCPB design stands out as a particularly matured and promising approach: current estimates present high values of TBR, it has none of the corrosion, erosion or MHD effect related issues that arise with liquid metal breeders, and makes use of a well-known, inert coolant (Helium). Improving shielding for more compact configurations and optimizing manufacturing and assembly of the design are key, as well as ensuring structural material longevity and accounting for permeation of tritium. Acknowledging this, foundations of extensive research and iterative design, within both the European fusion community and beyond, position it as one of the most viable options.

In any case, there are still persistent gaps in our understanding, notably in irradiation damage in SiC FICs for the DCLL design, tritium permeation pathways and manufacturing processes for intricate components such as WCCB's proposed 'envelopes' of ceramic breeding material. Addressing these will require integrated approaches: data from future high-flux fusion-relevant neutron sources, refined computational models and material investigation to ensure sustainable operation in fusion-relevant conditions.

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